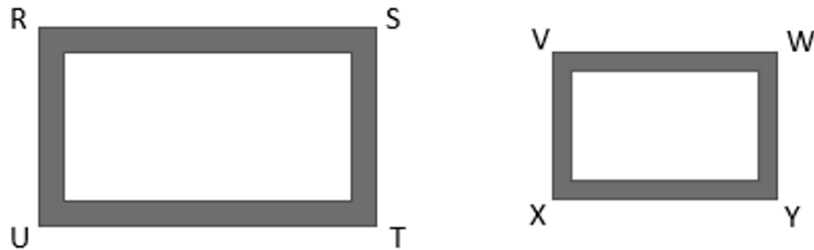


Hillary is custom building frames for a set of family photos. She has designed two similar rectangular frames, $RSTU$ and $VWYX$. The diagram shows the two different sized frames, where $WY = 12$ inches (in.), $VW = (3d + 11)$ in., $ST = 16$ in., and $RS = (8d - 4)$ in.



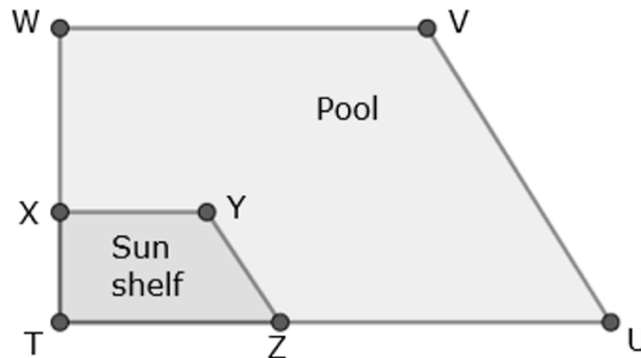
1. What is the value of d ?
2. What is the area of $RSTU$? Include units.
3. What is the perimeter of $VWYX$? Include units.

Hillary created a third similar rectangular frame, $LMNO$, shown below, where $LM = 12$ in.



4. What is the length of $\overline{LO}$? Include units.
5. What is the perimeter, in inches, of $LMNO$?

Laurel and her family are having a pool built in the backyard. The pool, $TUVW$, is shaped like a trapezoid, with a smaller similar trapezoid, $TZYX$, for a sun shelf. A diagram of the pool is shown, where $TU = (5a - 1)$ feet (ft), $TZ = (2a - 2)$ ft, $ZY = 5$ ft, $UV = 15$ ft, $VW = (10b + 1)$ ft, and $YX = (3b + 1)$ ft.



6. Determine the value of each variable:

$$a = \underline{\hspace{2cm}}$$

$$b = \underline{\hspace{2cm}}$$

7. Determine the value of each side:

$$WV = \underline{\hspace{2cm}}$$

$$TZ = \underline{\hspace{2cm}}$$

8. If $WV = WT$, what is the perimeter, in feet, of the pool?

Some measurements of angles of $TUVW$ and $TZYX$ are given.

$$m\angle VUZ = (12c - 6)^\circ, m\angle YZT = (10c + 5)^\circ, m\angle XTZ = 90^\circ,$$

$$m\angle TXY = (4d + 10)^\circ, m\angle TWV = (5d - 10)^\circ$$

9. Determine the value of each variable:

$$c = \underline{\hspace{2cm}}$$

$$d = \underline{\hspace{2cm}}$$

10. What is the measure of $\angle XYZ$?